

MathGov Core v12.6 and MHIOS v0.7 Independent Release Report

Core release: MathGov_Core_2026_09_v12.6_SGP_v8.5+2026.08.10.2

MHIOS release: MHIOS_v0_7+2026.08.10.2

Publication date: 10 August 2026

Review disposition: READY WITH BOUNDED OPEN WORK — CONCEPTUAL FREEZE
RECOMMENDED

Construct validity: UNTESTED

Executive finding

The five-stage MathGov architecture is coherent, necessary in its present separation, and sufficiently specified to freeze conceptually. The audit did not find a justified case for a sixth gate, an eighth scope or dimension, a TRC/CSV merger, weakened rights noncompensation, or a new authority layer.

The original attached builds were not yet freeze-ready because current release surfaces disagreed on citation metadata, build identity, and companion pins; the Canon overstated the meaning of a zero cell; a lock file and malformed text fragment were packaged; MHIOS had two live WDBIP v1.5 crosswalk references; and all governing tables had lost their professional style. These were bounded but genuine defects.

Builds MathGov_Core_2026_09_v12.6_SGP_v8.5+2026.08.10.2 and MHIOS_v0_7+2026.08.10.2 preserve those repairs and complete a further reality-grounding crosscheck without changing the governing cascade, formulas, thresholds, rights protections, SGP floors, or semantic versions. The newly supplied external engineering materials did not reveal a missing canonical layer. They did justify one informative consolidation: evidence at derivation, computation, implementation, projection, empirical test, operational qualification, and observed-outcome stages must not be promoted into a stronger later-stage claim. Core is ready for open research release; MHIOS is ready as a candidate implementation companion. Neither is empirically validated or deployment-authorizing.



1. Reconstructed architecture

RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS

RG is a claim-authority and reality-grounding precondition. RF/NCRC, TRC, and CSV are noncompensatory rejecting gates. RLS is a ranking layer operating only on the selectable set. This is one epistemic precondition, three qualification gates, and one post-qualification ranking stage, not five interchangeable scores.

Stage	Controlling question	Output boundary
RG/RSG	Are the material claims	Supported, bounded,

Stage	Controlling question	Output boundary
	adequately grounded for this decision and domain?	underdetermined, refused, or learning-action state.
RF/NCRC	Does the option preserve the noncompensable rights floor?	Rights-qualified or rejected/redesign/escalation.
TRC	Is catastrophic, ruinous, transformative, or irreversible tail exposure acceptable under the declared profile?	Tail-qualified, emergency-provisional, rejected, or refusal under deep uncertainty.
CSV	Can the option remain structurally viable under real dependencies, controls, authority, monitoring, and carried obligations?	Selectable, selectable with controls, not material, redesign, escalation, or failure.
RLS	Among only selectable options, which has the strongest disclosed 7 × 7 welfare profile?	Ordinal/comparative ranking with uncertainty, distributional, sensitivity, and decisiveness diagnostics.

The lifecycle continues beyond ranking: selection -> authorization -> exact action binding -> execution readiness -> monitored execution -> outcome observation -> requalification when material conditions change. Qualification is not authority, capability is not authority, authentication is not correctness, and a prior qualification is not a transferable permit.

2. Strongest verified features

- Rights and catastrophe protections are structurally non-dilutable by RLS benefit.
- RG exposes evidence domains, warrant domains, uncertainty, unknowns, boundary conditions, and refusal instead of laundering uncertainty through a score.
- TRC and CSV remain distinct: tail exposure is not the same question as structural governability and lifecycle viability.
- RLS cannot run before qualification or resurrect a rejected option.
- SGP separates moral-patienthood/protection, participation, capability, and authority and explicitly rejects consciousness-detector use.
- Exact action/snapshot/configuration binding closes the gap between an abstractly qualified option and the real action executed.
- Machine schemas, negative vectors, workbook surfaces, state registries, and hashes make a substantial portion of the architecture inspectable and replayable.

3. Genuine defects repaired

Defect	Severity	Consequence	Repair
Stale CITATION.cff and mixed earlier build metadata	High	Non-reproducible citation and dependency identity	Exact v12.6 / 10 August metadata, a distinct build 2026.08.10.2 identity, and hard verifier checks.
Active component maps and current-line prose contained old companion pins	High	Two valid implementers could choose different governing versions	Synchronized all active source and reading mirrors; added exact mirror checks.
MHIOS crosswalks still named WDBIP v1.5 and old MFDI/Source pins	High	False exact-compatibility claim	Corrected to WDBIP v1.6, MFDI v2.3, Source Coupling v2.3 and exact Core build.
Zero-cell biconditional was too strong	Medium	Offsetting subgroup/pathway harms could be misread as absence of change	Defined zero as net represented cell change and required non-masking disclosures.

Defect	Severity	Consequence	Repair
Normative coverage counted lines rather than modal clauses	Medium	Misleading coverage denominator and hidden obligations	154-clause census plus executable-check registry and explicit nonclaims.
Generic unstyled tables across governing DOCX files	Medium	Reduced scanability and professional usability	Restored restrained light-blue table system and verified 304 tables across Core and MHIOS.
Shipped office lock file	Medium	Dirty-source and tool-conflict risk	Removed and prohibited by verifier.
Accidental Canon table fragment	Low	Reader/converter confusion	Removed and synchronized mirrors.

4. Verification outcome

Surface	Final result
Core current pins, citation, and source hygiene	PASS
Core subordinate verification surfaces	9 / 9 PASS
Core run vectors	6 positive / 29 intended failures
SGP RMCP vectors	1 positive / 1 intended failure
WDBIP vectors	1 positive / 17 intended failures
Workbook	87 sheets / 1,643 formulas / 0 errors after live recalculation
Core reading surfaces	15 triples / 287 tables / 596 pages / 0 confirmed layout defects
MHIOS conformance	4 positive / 52 intended failures
MHIOS mutations	40 / 40 killed
MHIOS normative census	154 clauses; 65 any linkage; 55 direct; 89 unlinked
MHIOS document	17 tables / 66 pages / 0 confirmed layout defects

5. Reality-grounding and evidence-boundary crosscheck

The crosscheck was conducted as an independent requirements audit, not as a derivative design exercise. No external formula, primitive, ontology, taxonomy, verdict system, hardware stack, implementation logic, or proprietary term was imported.

Methodological challenge tested	Existing MathGov coverage	Build 2026.08.10.2 disposition
Separate derivation, computation, implementation, projection, measurement, and physical evidence.	MFDI v2.3 milestone claim discipline; PC-AEP v2.3 formal-conformance and specification-to-reality boundary; Scientific Maturity Ladder.	Retained the architecture and added one informative evidence-stage non-substitution matrix.
Freeze the evaluation projection before outcome-sensitive observation.	Canon projection pre-registration, parameter/configuration locking, MHIOS projection pre-registration record, independent replay packet.	Already covered; no new field or state added.
Allow a failed implementation, prediction, or physical test to propagate upstream.	MFDI dependency-localized failure propagation, non-retrofitting, re-derivation scope, distribution-shift and requalification protocol.	Already covered; verifier now protects the consolidated reporting surface.

Methodological challenge tested	Existing MathGov coverage	Build 2026.08.10.2 disposition
Distinguish a functioning implementation from domain validity, physical safety, or deployment readiness.	PC-AEP, MFDI, release claim boundaries, scientific maturity levels L0-L6.	Already normative/informative across current sources; made easier to inspect in one matrix.
Preserve monitoring, exact action binding, controls, preconditions, outcomes, and requalification at the execution boundary.	MHIOS action/specification/snapshot hashes, execution mandate and authority, readiness conjunction, outcome observation, and requalification.	Already covered; MHIOS report metrics and release verification were synchronized.
Keep reality external to the framework and refuse unsupported stronger claims.	RG/RSG, MFDI revision triggers, PC-AEP external evidence, Refusal, and release nonclaims.	Confirmed. Conceptual freeze remains recommended.

6. Changes deliberately rejected

- No sixth gate: no demonstrated failure remained that could not be resolved within RG, RF/NCRC, TRC, CSV, RLS, or the qualification-to-execution lifecycle.
- No TRC/CSV merger: merging would blur non-dilutable catastrophe exposure with control/governability and post-state viability.
- No cardinal moral-utility claim for RLS: the scoring system remains a governed comparative representation, not a measurement of intrinsic worth.
- No weakening of the rights floor or emergency benefit override.
- No universalization of configurable defaults as scientific constants.
- No SGP consciousness detector or capability-to-rights/authority shortcut.
- No AI normative judgment or execution authority.
- No semantic-version bump for corrections that preserve the external contract.

7. Remaining bounded open work

Internal coherence is not construct validity. The next program must test inter-rater reliability, parameter calibration, domain transfer, usability and burden, false confidence, institutional capture, legal validity, outcome quality, distribution shift, control effectiveness, and comparative performance against simpler baselines. High-consequence pilots should begin only after low-stakes retrospective replay and a no-AI MHIOS Minimal Vertical Slice establish that the workflow itself adds value.

8. Final disposition

READY WITH BOUNDED OPEN WORK — CONCEPTUAL FREEZE RECOMMENDED.

The corrected Core should now be frozen at the architectural level. Future changes should require new evidence, a demonstrated failure, or a genuinely new requirement. MHIOS should remain a versioned candidate companion under empirical development rather than being absorbed into Core authority.